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How Tight Are Physical Metals Markets? Supply, Demand, Balances, and Inventories Across Six Metals

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Abstract. The physical side of metals markets is tested against the data for copper, silver, gold, zinc, nickel, and tin, with shorter checks on lead and the platinum-group metals. The verdict is not uniform. Copper's 2025 refined market ran a surplus of 380 kt (preliminary, ICSG), even as the concentrate market hit its tightest terms on record: the 2026 term treatment charge benchmark settled at zero. Silver has posted six consecutive annual deficits, but the 2025 deficit was revised from a forecast 117.6 Moz to 40.3 Moz, and exchange stocks remain large enough that stock absorption, not acute scarcity, is the operative mechanism. Gold's published supply-demand balance is zero by construction, so no deficit thesis is testable; the price clears through roughly 222,600 tonnes of above-ground stocks. Zinc's 2025 balance of minus 33 kt is under 0.3 percent of consumption: close to balanced. Nickel moved from surplus into a forecast 2026 deficit of 32 kt, a forecast that itself moved 293 kt in six months. Tin shows small deficits on the thinnest data of the six. Mechanical 2030 scenarios flip sign across the four paths for every balance-testable metal, so no 2030 outcome is a forecast. The companion paper, Kilambi Research (2026), "Do Paper Markets Price Physical Scarcity?", KR-WP-2026-14, covers futures curves, positioning, delivery mechanics, and the macro decomposition; none of that is duplicated here. There was no pre-registered analysis plan; this is a retrospective evidence synthesis. The paper states, for each metal, the facts that would overturn its conclusions.

Keywords: physical markets, metal balances, inventories, incentive prices, supply scenarios, China, copper, silver, gold, zinc, nickel, tin

JEL Classification: Q31, Q33

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1 Introduction

Are physical metals markets tight? The question matters because the answer is routinely asserted from prices: high prices are read as scarcity, and scarcity is read as a structural deficit. This paper tests that chain metal by metal, using only the physical side of the ledger: mine and project supply, disaggregated demand, published balances, and visible inventories. Futures curves, positioning, delivery mechanics, and the macro decomposition of price moves are the subject of the companion paper, Kilambi Research (2026), "Do Paper Markets Price Physical Scarcity?", KR-WP-2026-14, and are not duplicated here.

The bottom line is that tightness is not a single condition and it is not uniform across metals. Copper's refined market was in surplus in 2025 while its concentrate market was the tightest on record. Silver has run deficits for six straight years, but the 2025 deficit shrank on revision and reported stocks are large. Gold cannot be in deficit under the accounting used to measure it. Zinc is close to balanced. Nickel was in surplus and may or may not flip in 2026. Tin is thinly supplied on weak data. None of the six presents an unambiguous case of acute physical scarcity, and for several the adverse evidence (falling utilization, revisions that shrink deficits, spot prices above incentive levels) points the other way.

Three design choices govern the paper. First, every material number carries its provenance: an evidence level (Observed, Reconstructed, Modelled, Assumed) and a source-quality grade (A, B, C), rendered as (Observed, A) and defined in Section 5. Second, the paper reports null and adverse findings alongside supportive ones; a scarcity thesis that cannot survive its own falsifiers is not stated as a conclusion. Third, the 2030 and 2035 scenarios are mechanical illustrations, not forecasts: four constant-growth-rate paths whose signs flip for every balance-testable metal, which is itself a finding about uncertainty.

The rest of the paper proceeds as follows. Section 2 sets out the institutional background needed to read physical balances and inventories. Section 3 places the exercise in the related literature. Section 4 describes the data and their vintages. Section 5 documents the methodology, including the scenario construction and the falsifier framework. Section 6 reports results metal by metal and then across metals. Section 7 states the limitations, the known gaps, and the robustness checks, including what would overturn each conclusion. Section 8 concludes. Section 9 lists references. The data appendix inventories the machine-readable dataset that backs every figure and table.

2 Institutional background

A refined-metal balance is production minus consumption over a calendar year, as published by the study groups (ICSG for copper, ILZSG for zinc and lead, INSG for nickel, ITA for tin). Consumption in these accounts is apparent consumption (production plus net trade plus stock changes), not surveyed end use, so a balance near zero can hide regional or timing mismatches. Balances are published with a lag and revised: copper's 2024 preliminary surplus was first reported at 301 kt and later revised to 69 kt (Observed, B) (International Copper Study Group, 2026), a 232 kt revision that disciplines how much weight any single vintage can carry.

Gold is the exception that proves the rule about balance accounting. The World Gold Council's supply and demand series are constructed so that the two sides sum to the same total: total supply equals total demand by construction, and the published balance is zero every year (Observed, A)

(World Gold Council, 2026). No physical deficit or surplus thesis is testable under that accounting. Gold analysis therefore has to work through stocks (above-ground inventories, central-bank holdings, ETF positions) rather than flows.

Silver sits between the two conventions. The World Silver Survey publishes a market balance (supply minus demand) that can be negative, and it has been negative for six consecutive years through the 2026 forecast (OBSERVED, B) (Metals Focus, 2026). But the survey's demand includes categories (notably investment and stock changes) whose measurement is weaker than mine-supply measurement, and its forecasts have been revised by large amounts, as Section 6.2 documents.

Exchange inventories are the visible margin of the physical market, but they are a subset of total stocks. The LME reports stocks by warehouse with an on-warrant/off-warrant split; only on-warrant metal is immediately deliverable. COMEX reports registered metal (warranted for delivery) separately from eligible metal (meets specifications but not committed). SHFE reports bonded and exchange stocks. Reported days of cover divide a stock level by a consumption rate, so the same tonnage reads differently depending on the denominator year chosen; all days-of-cover figures here use 2025 or 2026 consumption as noted. Inventories also double count across pools when summed naively (for example, bonded stocks that are also reported in national totals), so combined figures are shown separately and flagged as non-additive.

Treatment and refining charges (TC/RC) are the smelter's margin for converting concentrate into refined metal, negotiated annually as a benchmark and traded daily on the spot market. When concentrate is scarce relative to smelting capacity, TC/RCs fall; the smelter pays more for feed. A zero or negative TC means the smelter processes concentrate for no fee or pays the miner, the textbook signature of upstream tightness combined with downstream overcapacity. This is a concentrate-market signal, not a refined-metal signal, and the two can point in opposite directions, as copper in 2025-2026 demonstrates.

The incentive price is the price level at which new supply is expected to be sanctioned: the full-cycle cost of the marginal project plus a required return. It is an industry construct, estimated from project cost curves (Wood Mackenzie, S&P Global, company filings), and it is reported here as a range, not a point, because cost curves move with input prices, royalties, and financing terms (RECONSTRUCTED, C). A spot price persistently above the incentive range should, with a lag, call forth supply; a spot price below it should eventually curtail it. Where no defensible incentive estimate was found, none is quoted: gold is the case, and Section 6.3 explains why.

China's role is structural, not cyclical, in this paper's window. China accounts for roughly half of world refining in copper, zinc, and tin and a large share of consumption, while its mine share is much smaller (Table 1). That wedge makes the rest of the world dependent on Chinese smelting and refining capacity and makes Chinese concentrate imports, premiums, and stockpiling policy (the State Reserve Bureau, SRB) first-order variables for physical tightness. SRB stockpile levels are not published and remain an Assumed-C input throughout.

Secondary supply (recycling) responds to price with a lag and to collection infrastructure with a longer one. Lead is the extreme case, with recycling supplying roughly 60 to 70 percent of refined metal (RECONSTRUCTED, B) (International Lead and Zinc Study Group, 2026); copper, zinc, and tin recycling shares are less well measured and are flagged as gaps in Section 7. Recycling is a supply response, but it is not a mine, and it cannot be scaled by price alone where collection is the constraint.

Energy is a smelting cost, and in 2022 it became a smelting constraint. European zinc smelters curtailed output when observed electricity prices reached roughly EUR 340 to 527/MWh (OBSERVED,

B) (Nyrstar et al., 2022). That range is what was observed during curtailments; it is not a proven break-even threshold, and the paper does not present it as one. The episode matters because it shows how a power-price shock converts into a physical supply shock without any change in mine output.

Finally, the boundary with the companion paper: futures curve shapes (contango, backwardation), lease rates, physical premiums, delivery and warrant mechanics, CFTC positioning, and the regression decomposition of price moves into macro factors are all in Kilambi Research (2026), "Do Paper Markets Price Physical Scarcity?", KR-WP-2026-14. This paper cites that boundary once and does not reproduce it. Where a physical conclusion depends on a paper-market fact (for example, that a futures curve does not price nearby scarcity), the companion paper is the authority and the claim is flagged accordingly.

3 Related literature

The economics of exhaustible resources begins with Hotelling (1931), whose rule links the price path of a depletable stock to the interest rate. The rule's empirical record in metals is weak, which is itself informative: above-ground stocks, recycling, substitution, and new discoveries break the fixed-stock assumption, and this paper's gold section is in effect a documentation of how completely stocks dominate flows for one metal.

The theory of storage (Fama and French, 1987) connects inventory levels to the spread between spot and futures prices: low inventories should coincide with backwardation (spot above futures) as the convenience yield of holding metal rises. The physical half of that relationship, inventories and balances, is this paper's subject; the price half is the companion paper's. The two papers are designed to be read as a pair precisely because storage theory is a joint hypothesis about both.

Pindyck (2001) models commodity price dynamics with mean reversion and stochastic volatility, and Cashin and McDermott (2002) document small long-run trends with large variability across commodities. Both counsel humility about extrapolating recent price levels, a caution this paper operationalizes through scenario ranges that flip sign rather than point forecasts. Slade (1982) finds cycles in natural-resource prices in the frequency domain; the balance histories in Section 6 show the corresponding quantity cycles, including copper's swing from deficit to surplus within five years.

On identification, Kilian (2009) disentangles demand and supply shocks in oil markets and shows that not all price increases are supply stories. The paper applies the same discipline qualitatively: a rising price with a rising balance and rising stocks is a demand or financial story until proven otherwise, and several metals in Section 6 fit that description.

On market structure, Tilton (1999) argues that the long-run availability of nonfuel minerals is governed more by technology, exploration, and institutions than by fixed geological limits, while Geman (2005) systematizes the pricing of commodity derivatives including the delivery and storage mechanics this paper excludes by design. Humphreys (2015) documents the remaking of the mining industry around Chinese demand and the concentration of processing capacity that Section 6.9 measures.

What this paper adds is a forensic, balance-by-balance and stock-by-stock accounting across six metals on a common evidence standard, with stated falsifiers. The literature above largely works from prices; this paper works from quantities and asks whether the quantities support the stories the prices are sometimes used to tell.

4 Data

The dataset is a machine-readable compilation of published balances, inventories, prices, project pipelines, and China shares for copper, silver, gold, zinc, nickel, tin, lead, and the platinum-group metals, assembled September 2026. Every row carries a source, an as-of date, an access date, and an evidence tag. The data appendix (Section 10) inventories the files. The institutional sources are the study groups and industry bodies: U.S. Geological Survey (2026), International Copper Study Group (2026), International Lead and Zinc Study Group (2026), International Nickel Study Group (2026), International Tin Association (2025), World Gold Council (2026), Metals Focus (2026), International Energy Agency (2026), World Platinum Investment Council (2026), and Johnson Matthey (2026). Spot prices are August 2026 monthly averages from the IMF Primary Commodity Price System (International Monetary Fund, 2026); exchange stocks are LME, COMEX, and SHFE public reports (Kilambi Research, 2026a).

Table 1 summarizes the latest observation per metal. The table is the paper's summary-statistics requirement and should be read with two cautions stated once here and applied throughout: vintages differ across cells (a September 2026 stock snapshot sits next to a February 2026 balance vintage), and preliminary balances are revised, sometimes by amounts larger than the balance itself.

Table 1. Summary statistics: latest balance, forecast, visible stocks, price, and China share. The evidence column gives grades (Section 5 vocabulary) for the five data columns in order: balance, forecast, stocks, spot, China share. Class letters: O = Observed, R = Reconstructed, M = Modelled, A = Assumed; "–" marks not applicable.

Metal	Latest balance	2026 forecast	Visible stock (days)	Spot, Aug 2026	China ref. share	Evidence
Copper	+380 kt, 2025p	-150 kt (stale)	13.1	$14,331/t	49%	OC / OC / RB / OA / RB
Silver	-40.3 Moz, 2025	-46.3 Moz	103.5 / 30.5 (reg.)	$65.16/oz	n.a.	OC / OC / OB / OA / –
Gold	0 t (by constr.)	n.a.	60 yrs (above gr.)	$4,418.5/oz	n.a.	OA / – / RB / OA / –
Zinc	-33 kt, 2025p	-19 kt	3.2 / 19.5 (ILZSG)	$3,874.5/t	47%	OB / OC / OB / OA / RB
Nickel	+283 kt, 2025p	-32 kt	46.6	$16,751/t	29%	OB / OC / RB / OA / RA
Tin	-5.6 kt, 2025F	n.a.	9.2	$55,385/t	52%	OC / – / RB / OA / RA
Lead	+82 kt, 2025F	+82 kt	10.1	$2,244/t	38%	OC / OC / OB / OA / OC
Platinum	-1,082 koz, 2025	-240/-371 koz	n.a.	$1,980/oz	n.a.	OB / OC / – / OA / –

Notes: p = preliminary; F = forecast; reg. = COMEX registered; n.a. = not located or not applicable; platinum 2026 forecast is WPIC/JM. Sources: International Copper Study Group (2026), Metals Focus (2026), World Gold Council (2026), International Lead and Zinc Study Group (2026), International Nickel Study Group (2026), International Tin Association (2025), World Platinum Investment Council (2026), International Monetary Fund (2026), Kilambi Research (2026a).

Three measurement problems recur and are handled explicitly. First, apparent consumption (used in balance denominators) can diverge from true end use when unreported stocks move; Chinese bonded copper stocks, currently unknown, are the largest such blind spot (ASSUMED, C). Second, inventory

snapshots mix dates and some pools overlap, so combined stock figures are shown separately from exchange-deliverable figures and never summed silently. Third, end-use splits for copper, zinc, nickel, and tin are weak reconstructions (RECONSTRUCTED, C); the paper uses them only to sign demand direction, never to size a balance.

5 Methodology

Balance accounting. The annual refined balance is production minus apparent consumption as published by the relevant study group. For copper, zinc, nickel, lead, and tin this is a testable quantity with a published history. For silver the World Silver Survey balance is reported and testable but rests on weaker demand measurement. For gold the WGC balance is zero by construction and no scenario is run on it; scenario variation for gold is in the composition of above-ground supply, discussed qualitatively.

Days of cover. A stock level divided by a daily consumption rate (annual consumption divided by 365). The paper reports days of cover for exchange-deliverable pools (LME on-warrant or total, COMEX registered, SHFE) and, separately, for broader reported pools (ILZSG totals, combined exchange sums). The two are not interchangeable and the text never treats them as such.

Incentive prices. Reported as ranges compiled from industry cost-curve work (Wood Mackenzie, 2025; SMM and CRU, 2026), (RECONSTRUCTED, C). They are hurdle rates for sanctioning new supply, not price forecasts. The paper compares spot to the incentive range as a consistency check: sustained spot above the range without a supply response is evidence against a simple scarcity story, not for one.

Project pipeline. Named projects carry a capacity, an earliest year, and a probability weight reflecting status, financing, permitting, and jurisdiction risk. Only probability-weighted capacity is ever quoted; announced (nameplate) capacity is not used, because unweighted pipeline totals systematically overstate deliverable supply. Weights are (MODELLED, B) or (MODELLED, C) and the paper keeps them visible rather than hiding them inside a total.

Scenarios. Four mechanical paths to 2030 and 2035: conservative (low demand growth, high supply response), base (central), high demand, and low supply. Each applies constant growth-rate assumptions to supply and demand components; they are illustrations of sensitivity, (MODELLED, C), not forecasts, and the paper assigns them no probabilities. No empirically calibrated scenario probabilities were available, so no expected-value balance is reported. The four paths are mapped to plain-language labels using the 2030 balance as a share of consumption: within about ± 1 percent, adequately supplied; a deficit of 1 to about 5 percent, modest deficit; about 5 to about 10 percent, severe structural deficit; above about 10 percent, or implied stock exhaustion, acute squeeze. The labels are the authors' descriptive reading of mechanical paths, kept only where the mapping does not mislead (gold is excluded).

Falsifiers. For each metal the paper states observable facts that would overturn its conclusion, with approximate thresholds where the data allow. Falsifiers are the paper's pre-commitment device in the absence of a pre-registered plan: they say what evidence would change the verdict, in advance.

Evidence vocabulary. Every material number carries (OBSERVED, A/B/C), (RECONSTRUCTED, A/B/C), (MODELLED, A/B/C), or (ASSUMED, A/B/C). Observed means a published measurement; Reconstructed means computed from published components; Modelled means produced by an explicit model; Assumed means a judgment input. A, B, C grade source quality from primary and audited (A) through secondary and industry press (B) to single-source or weak (C). Unknown stays unknown: the paper marks gaps rather than filling them.

Retrospective disclosure. There was no pre-registered analysis plan. This is a retrospective synthesis of published data compiled September 2026. The falsifier framework and the both-directions robustness checks in Section 7 are the substitute discipline. Production note: this paper was produced by Kilambi Research's quantitative research pipeline with AI assistance; all analysis was verified against source data.

6 Results

6.1 Copper: a loose annual market on top of a tight concentrate market

Copper's refined balance ran deficits of 484 kt, 475 kt, 431 kt, and 87 kt from 2020 through 2023, then flipped to preliminary surpluses of 69 kt in 2024 and 380 kt in 2025 (OBSERVED, C) (International Copper Study Group, 2026). The October 2025 forecast for 2026 was a 150 kt deficit, but that vintage predates the 2025 actual and is stale (OBSERVED, C). The annual market, on the published numbers, is loose: two consecutive surpluses, the second the largest in the six-year window (Figure 1).

The upstream market tells the opposite story. The annual term treatment-charge benchmark fell from \$80/dmt for 2024 to \$21.25/dmt for 2025 and settled at \$0/dmt for 2026, the first zero benchmark on record (Antofagasta-Jiangxi Copper), and spot TCs printed at negative \$146.15/dmt in July 2026 (OBSERVED, B) (Reuters, 2026) (Figure 2). Smelters are processing concentrate for nothing or paying miners to take it: the textbook signature of concentrate scarcity against smelting overcapacity. China, with about 49 percent of world refining (RECONSTRUCTED, B) and 58 percent of consumption (OBSERVED, A), built the overcapacity; its smelters' import dependence for feed is roughly 86 percent (ASSUMED, C), which is why the TC collapse is concentrated there.

The incentive check argues against a refined-scarcity reading. Spot was \$14,331/t in August 2026 (OBSERVED, A), above the estimated incentive range of \$9,370 to \$11,000/t (RECONSTRUCTED, C) (Wood Mackenzie, 2025) (Figure 9). A market priced above its incentive level for an extended period should be calling forth supply, and the pipeline, probability-weighted, is responding: about 2.2 Mt of weighted additions across 29 tracked projects, 1.7 Mt of it with earliest years at or before 2030, led by the Grasberg recovery (280 kt weighted), the Collahuasi recovery (146 kt), and Reko Diq Phase 1 (140 kt) (MODELLED, B/C) (Figure 3). Against that stands cumulative depletion of roughly 4.6 Mt over 2026-2035 (MODELLED, C), an estimate whose underlying S&P data were not retrieved and which is carried with a wide confidence band.

Visible refined stocks do not confirm refined tightness either. LME on-warrant stocks were about 1.6 days of cover, SHFE about 1.0 day, LME total 3.2 days, and COMEX 8.9 days; combined LME plus COMEX plus SHFE was 1,027 kt or 13.1 days (RECONSTRUCTED, B) (Kilambi Research, 2026a) (Figure 4). Exchange cover is thin on the LME prompt specifically, which is consistent with regional tightness, while the global exchange pool is not depleted.

The mechanical scenarios span a 2030 balance from +1,044 kt (conservative) to -1,772 kt (high demand), with the base at -284 kt (-0.9 percent of consumption) (MODELLED, C). The sign flips across paths; the base path maps to "adequately supplied" under the Section 5 thresholds, the conservative path likewise, and the high-demand and low-supply paths to "severe structural deficit" and "modest deficit" respectively. None of these is a forecast. The honest reading is that copper's 2030 balance is sign-ambiguous under small changes in growth assumptions, which is why the paper keeps all four paths visible instead of selecting one.

What would overturn the verdict: a third consecutive refined surplus would confirm looseness rather than overturn it; the overturning facts would be LME on-warrant stocks rebuilding above 200 kt alongside a return to contango, term TC/RC recovering above about $80/dmt, or project supply failing to replace depletion such that mine output falls two years running. On current evidence the annual refined market is adequately supplied and the tightness is upstream and regional.

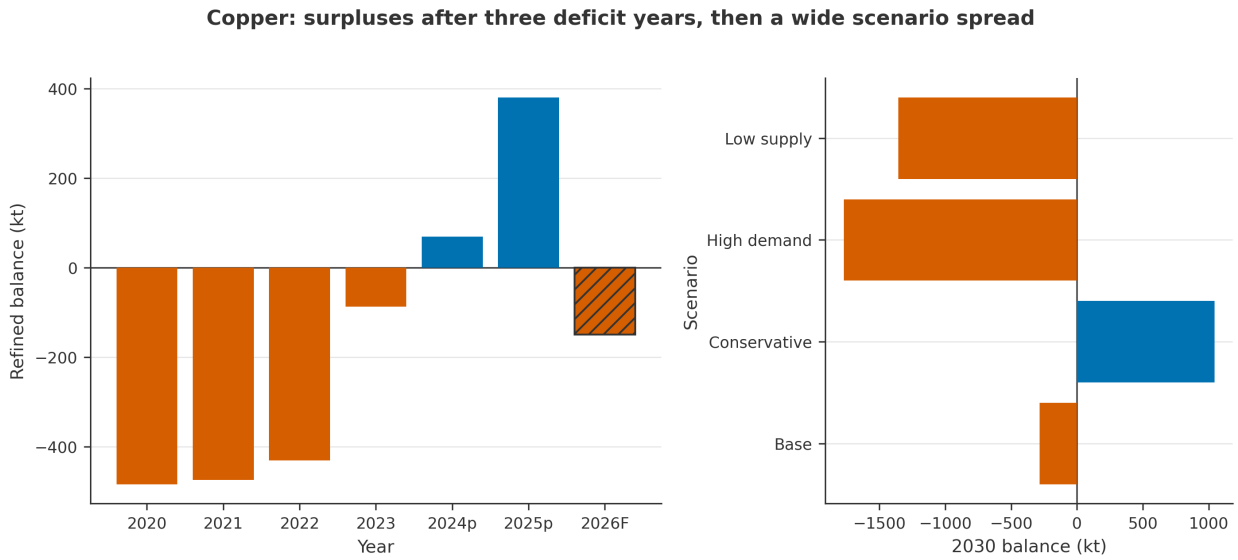


Figure 1. Copper balances: surpluses in 2024-2025 after three deficit years; 2030 scenarios flip sign. Left: ICSG refined balances 2020-2026 (2026 is the October 2025 forecast vintage, now stale). Right: mechanical 2030 scenarios, all four paths kept (Modelled, C).

6.2 Silver: persistent deficits, shrinking on revision, absorbed by stocks

Silver's market balance was +45.0 Moz in 2020, then -79.3, -249.6, -200.7, and -149.0 Moz from 2021 through 2024, -40.3 Moz in 2025, with -46.3 Moz forecast for 2026: six consecutive deficit years (OBSERVED, B/C) (Metals Focus, 2026) (Figure 5). The bottom line comes first, but the caveat belongs in the same sentence: the 2025 deficit was first forecast at -117.6 Moz and later reported at -40.3 Moz, a 77.3 Moz revision, and the 2024 figure carries the same survey methodology. Deficits this persistent would, in a textbook market, have broken the price years ago; silver's price did

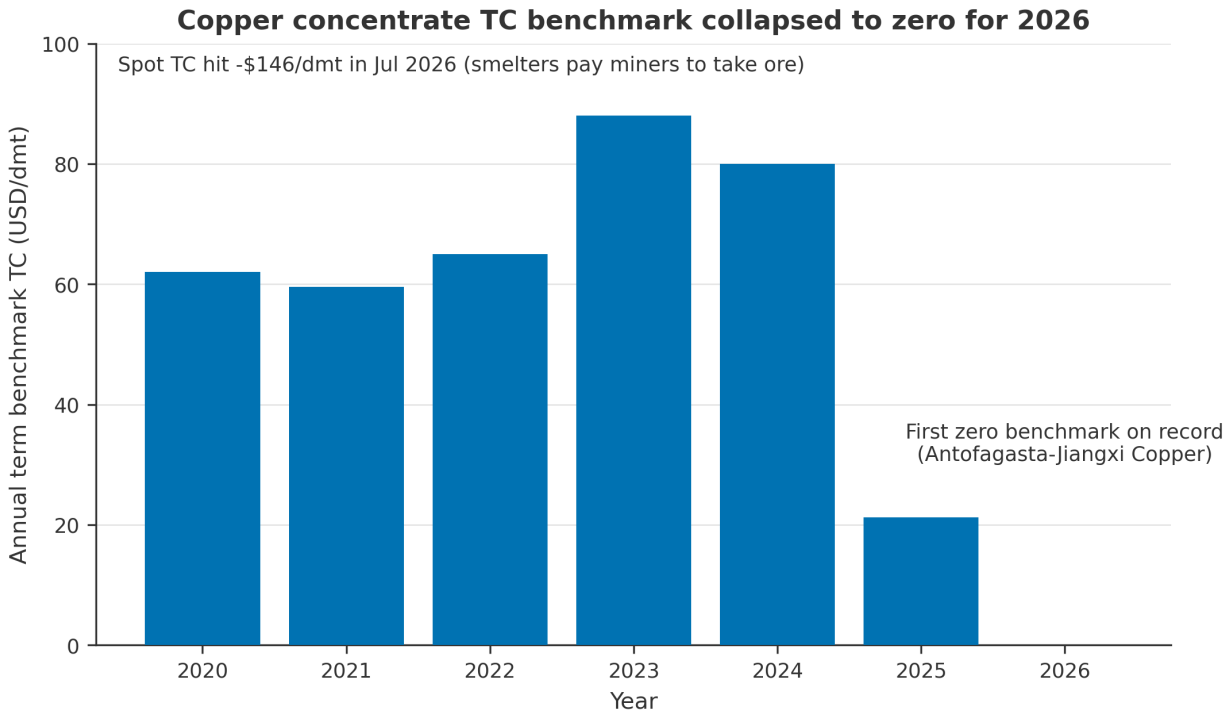


Figure 2. Copper concentrate term benchmark collapsed to zero for 2026. Annual term treatment charge, USD/dmt; spot TC hit -$146/dmt in July 2026. Source: Reuters reporting (Observed, B).

rise, but the multi-year coexistence of large reported deficits with orderly markets is economically unresolved and is stated as such rather than explained away.

Stocks are the absorption mechanism. COMEX total stocks were 330 Moz (103.5 days) with 97.3 Moz registered (30.5 days) (Observed, B) (Kilambi Research, 2026a) (Figure 4). The registered category, the deliverable margin, is the binding one; eligible stocks are a weaker buffer. Reported deficits over 2021-2025 sum to roughly 719 Moz against those pools, which implies either unreported stock draws, demand overstatement, or both. The paper does not adjudicate the split; it records that the published flows and the published stocks do not reconcile without one.

Supply structure is the reason deficits do not self-correct quickly. About 72 to 74 percent of mine supply is a by-product of lead, zinc, copper, and gold mines (Observed, B) (Metals Focus, 2026); 9 of the top 20 silver mines are primarily copper, zinc, or lead operations contributing about 145 of 279 Moz (Figure 6). A silver price signal largely cannot call forth silver supply because the investment decision belongs to another metal's economics: the by-product trap. Recycling is the elastic margin that did respond: 197.6 Moz in 2025, forecast 211.3 Moz for 2026 (Observed, B), up on the price.

The incentive comparison is mixed. Spot was $65.16/oz in August 2026 (Observed, A); primary-mine all-in costs plus a hurdle sit around $25-30/oz and marginal greenfield needs roughly $35-45/oz (Reconstructed, C) (Figure 9). Spot is well above both, which should incentivize primary projects, but the by-product structure means the marginal silver ounce is not the marginal investment decision. Demand is concentrated in industrial uses with photovoltaic demand the swing factor (Reconstructed, C) (International Energy Agency, 2026).

Scenarios give a 2030 base balance of -12.5 Moz (-1.1 percent), inside "adequately supplied", with a

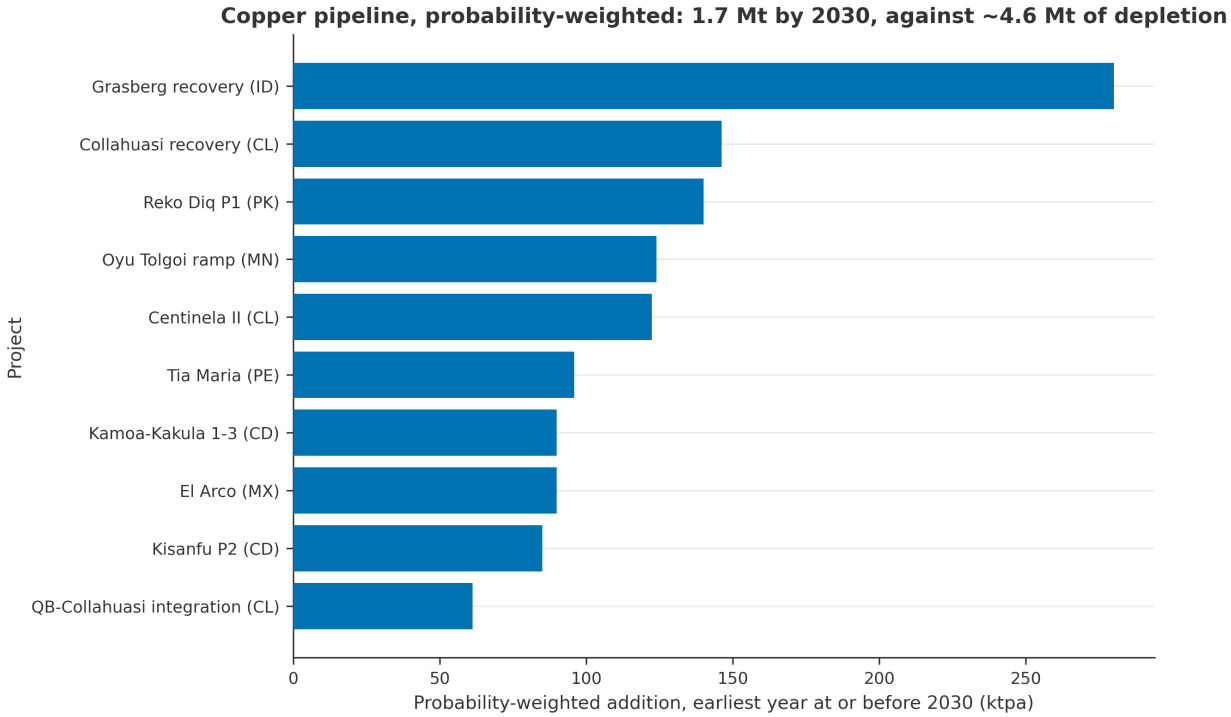


Figure 3. Copper pipeline, probability-weighted: 1.7 Mt of additions by 2030 against about 4.6 Mt of depletion. Weights reflect status, financing, permitting, and jurisdiction risk (Modelled, B/C).

range from +86.6 to -106.9 Moz (Modelled, C). The verdict: persistent flow deficits, real, but absorbed by stocks and subject to large revisions; acute scarcity is not established. What would overturn it: two consecutive annual surpluses with spot below $35/oz, COMEX registered stocks rebuilding above 150 Moz, vault free float rising above 40 percent, or photovoltaic silver demand falling below about 130 Moz. Conversely, a registered-stock draw below 20 days of cover with lease-market stress would confirm the tight reading; that is the companion paper's evidence to weigh (Kilambi Research, 2026b).

6.3 Gold: the balance is zero by construction; the price clears through stocks

The World Gold Council's supply and demand both total about 5,002 t for 2025, and the published balance is zero, as it is every year, because the accounts are constructed to balance (Observed, A) (World Gold Council, 2026). This is the single most important fact about gold in this paper: no deficit or surplus thesis is testable under WGC accounting, and any commentary that cites a gold "deficit" is misreading the table. The paper therefore works through demand composition and stocks.

Mine supply was 3,672 t in 2025 (Observed, A); recycling reached 1,404.3 t, up 3 percent on 2024 (Observed, B). Demand composition shifted hard in 2025: investment demand (bars, coins, ETFs) reached 2,175.3 t, up 84 percent, with ETF inflows of 801 t taking holdings to a record 4,025 t; jewelry fell to 1,542 t from 1,877 t; central banks bought 863.3 t, down from the revised 1,092.4 t in 2024; technology took 323 t (Observed, A/B) (World Gold Council, 2026) (Figure 7). The 2025 rally was investment-led while jewelry and official-sector demand stepped back: a financial bid, not

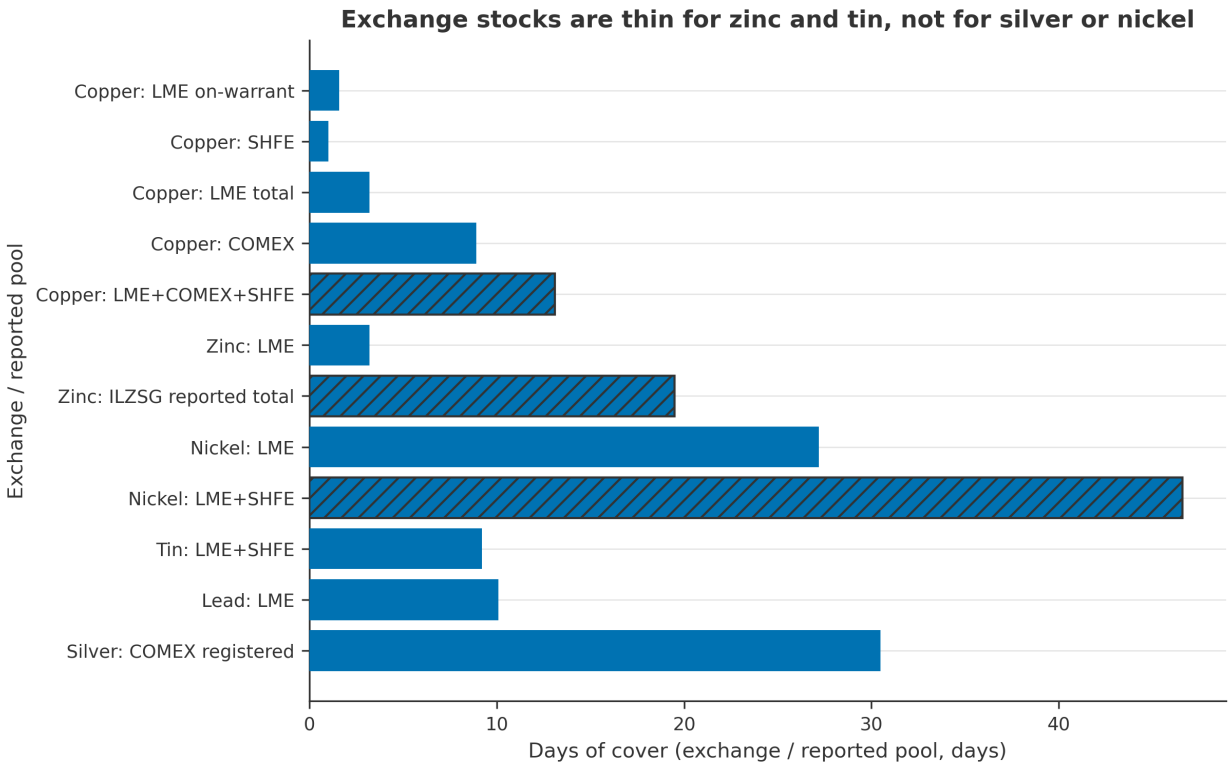


Figure 4. Exchange stocks are thin for zinc and tin, not for silver or nickel. Days of cover for exchange-deliverable pools (solid) and combined or reported pools (hatched; not all deliverable, pools may overlap). Sources: LME, COMEX, SHFE, ILZSG.

a fabrication squeeze.

Above-ground stocks are about 222,600 t, roughly 60 years of mine supply (Reconstructed, B). That stock-to-flow ratio is why gold's price can move hundreds of dollars on portfolio shifts while mine output barely matters: the clearing mechanism is revaluation of the stock, and the marginal suppliers of metal to the market are holders, not miners. No defensible incentive price was found for gold and none is quoted; a cost curve for the marginal gold project exists in industry databases, but the paper could not verify a figure it was willing to stand behind, and inventing one would be worse than the gap.

Central-bank buying, at 863.3 t in 2025 and forecast about 850 t for 2026 (Observed, A), remains the structural flow support, roughly a quarter of mine supply. China is the largest miner at about 380 t (11.5 percent of world mine output) (Observed, A) (U.S. Geological Survey, 2026), and the PBoC holds about 2,280 t (Observed, A).

No 2030 scenario is run for gold: the balance is identically zero on every path. Scenario variation would be in the composition of above-ground supply (ETF versus central-bank versus jewelry stocks), which the paper does not model. What would overturn the verdict: central-bank buying below 500 t for two consecutive years combined with mine-supply growth above 2 percent for three years and recycling above 1,600 t would establish a flow-loose market; conversely the current verdict needs no overturning because it makes no scarcity claim. Gold is the metal where the physical-tightness question is least well posed, and the paper leaves it unposed.

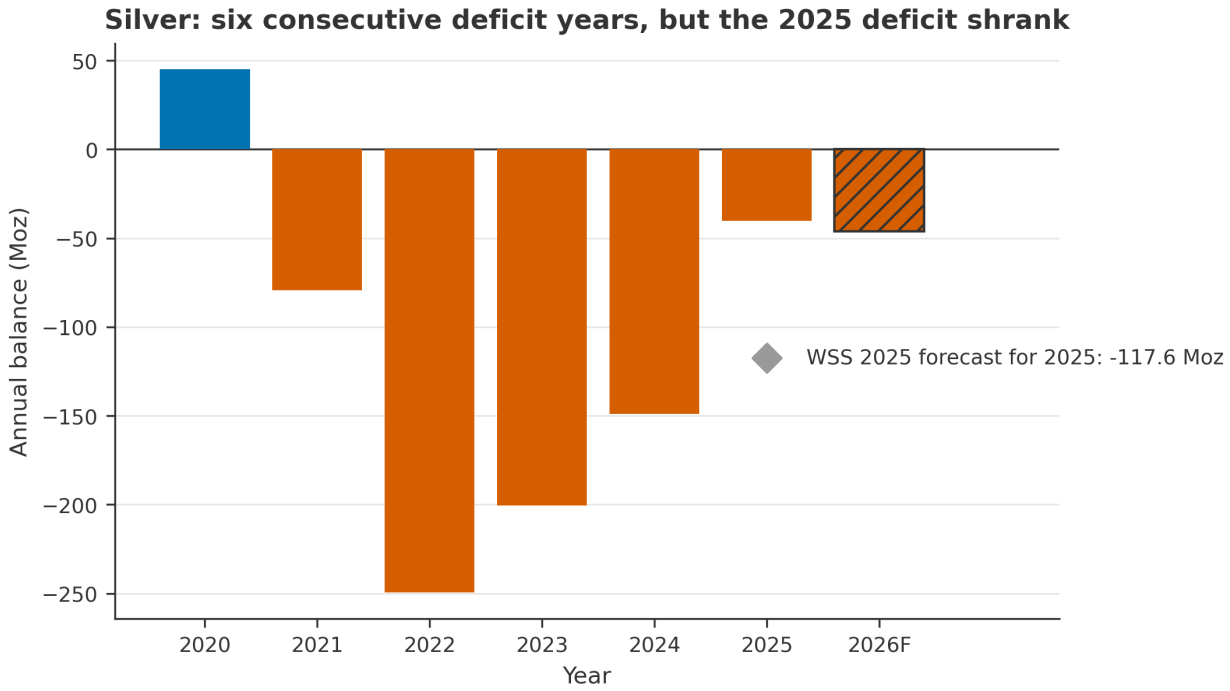


Figure 5. Silver: six consecutive deficit years, but the 2025 deficit was revised from a forecast 117.6 Moz to 40.3 Moz. Diamond marks the superseded WSS 2025 forecast for 2025. Source: World Silver Survey (Observed, B/C).

6.4 Zinc: close to balanced; the constraint is concentrate, not smelters

Zinc's refined balance was +536 kt in 2020, -144 kt in 2021, +205 kt in 2023, -69 kt preliminary in 2024, -33 kt preliminary in 2025, and -19 kt forecast for 2026 (Observed, B/C) (International Lead and Zinc Study Group, 2026) (Figure 8). The 2022 figure was not located and is shown as a gap, not interpolated. The 2025 deficit of 33 kt is under 0.3 percent of consumption: the market is close to balanced, and the bottom line is that small. A 2024 relay conflict (-69 kt versus -62 kt across sources) is disclosed and does not change the picture at these magnitudes.

Stocks tell a two-pool story. LME stocks were 123 kt, 3.2 days of cover: thin on the exchange (Observed, B). ILZSG reported total stocks were 739 kt, 19.5 days: comfortable in aggregate (Observed, B) (Kilambi Research, 2026a; International Lead and Zinc Study Group, 2026) (Figure 4). The wedge between the two is the paper's recurring inventory lesson: exchange tightness and system looseness coexist, and quoting one without the other misleads.

The structural fact is upstream. China holds about 34 percent of mine output and 47 percent of refining (Reconstructed, B), and smelters' bargaining position is weak: Spot TCs went negative in Q4 2024 (minus $40/dmt, SMM) and were assessed at minus $113/dmt in August/September 2025 (SMM via Andy Home); Chinese buying guidance for Q2 2026 was $35-70/dmt, down from $105-120/dmt for Q1 2026; the annual benchmark (Korea Zinc/Teck) fell from $165/dmt for 2024 to $80/dmt for 2025, the lowest in more than 50 years per CRU, and about $85/dmt for 2026 (Observed, B) (SMM and CRU, 2026). Concentrate availability, not smelter capacity, is the current constraint. The 2022 European curtailments are the energy-sensitivity case study: smelters cut output with observed electricity prices around EUR 340 to 527/MWh (Observed, B) (Nyrstar

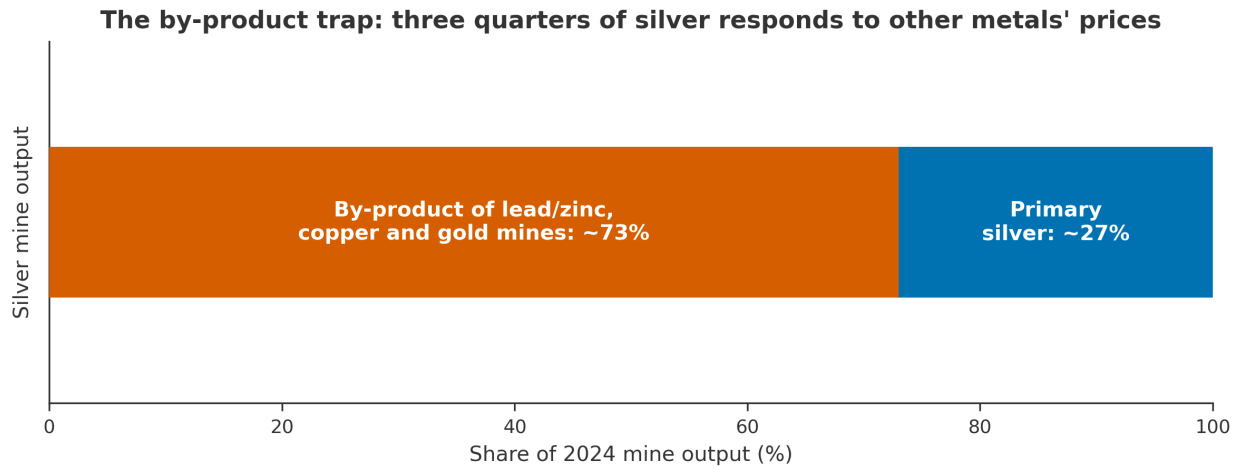


Figure 6. The by-product trap: about three quarters of silver mine output responds to other metals' prices. 9 of the top-20 silver mines are copper, zinc, or lead operations. Source: WSS 2026 (Observed, B).

et al., 2022), a range the paper reports as observed during curtailments, not as a break-even threshold. Restarts since (Nyrstar Budel, Glencore Nordenham) sit against brownfield restart economics of roughly $2,200-3,000/t, with top-tier greenfield all-in costs as low as $1,345/t at Tala Hamza (Reconstructed, C); spot at $3,874.5/t in August 2026 (Observed, A) clears those hurdles (Figure 9).

Scenarios give a 2030 base balance of -10 kt (-0.07 percent): adequately supplied, with a range of +743 to -599 kt (Modelled, C). The verdict is near-balance with upstream fragility. What would overturn it: ILZSG reported stocks above 850 kt, LME stocks rebuilding above 200 kt, European smelter restarts completing, or concentrate TCs recovering materially. A large mine disruption (Red Dog, Rampura Agucha, or Antamina scale) would tighten the concentrate market further without changing the refined verdict much in the first year.

6.5 Nickel: surplus behind it, a forecast flip ahead, and a two-tier price structure

Nickel's balance was +108 kt in 2020, -166.6 kt in 2021, +112.2 kt in 2022, +238.8 kt in 2023, +112 kt in 2024, +283 kt preliminary in 2025, with -32 kt forecast for 2026 (Observed, B/C) (International Nickel Study Group, 2026) (Figure 8). The recent market was in surplus; the forward look is a small deficit. But the 2026 forecast moved from +261 kt to -32 kt within six months, a 293 kt revision that is nine times the forecast balance itself. A forecast that unstable carries almost no weight, and the paper treats the 2026 figure as illustrative rather than informative.

Indonesia supplied about 67 percent of 2025 mine output (Observed, B), the highest single-country concentration among the six metals, and its ore-export and quota policies are the swing factor for world supply. LME plus SHFE stocks were 478 kt, 46.6 days of cover (Reconstructed, B) (Kilambi Research, 2026a): the deepest exchange cover of any base metal here (Figure 4). Nothing about the stock position suggests tightness.

The price structure is two-tier. Indonesian nickel pig iron cash costs sit around $8,850/t; Western sulphide and HPAL projects need roughly $18,000/t or more (Reconstructed, C). Spot at $16,751/t in August 2026 (Observed, A) sits between the two (Figure 9): profitable for the

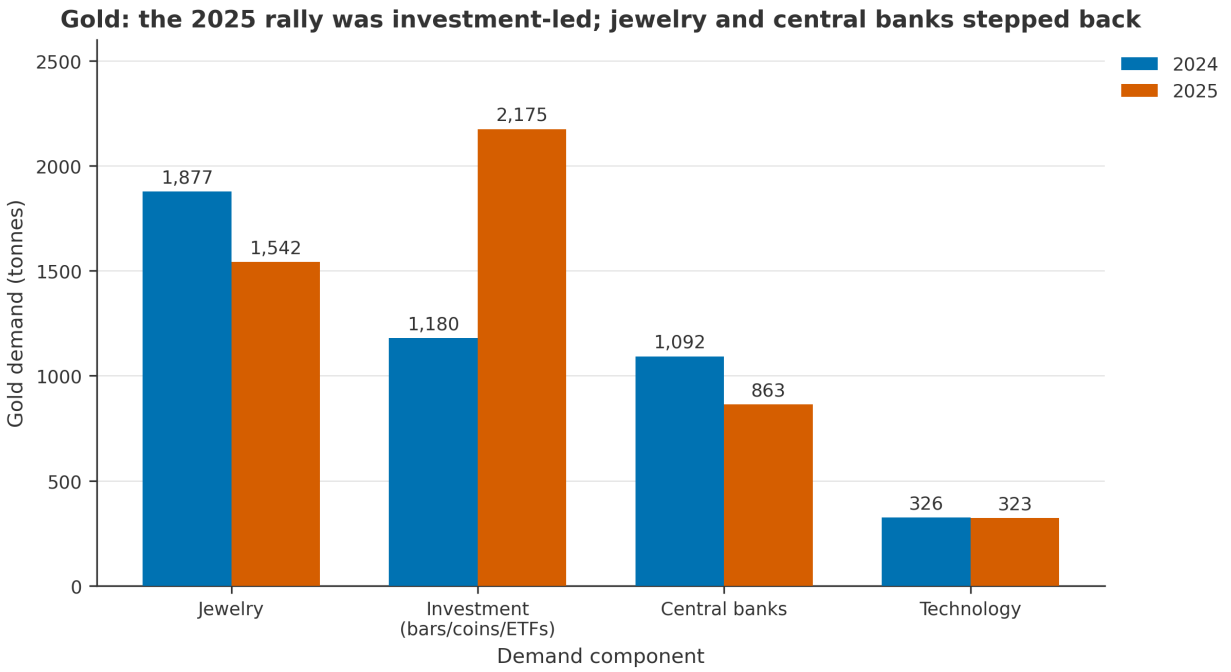


Figure 7. Gold: the 2025 rally was investment-led; jewelry and central banks stepped back. Demand components in tonnes, 2024 versus 2025. Source: World Gold Council (Observed, A/B).

Indonesian cost curve, below the Western incentive. That wedge explains both the investment strike outside Indonesia and the continued Indonesian supply growth; it is a cost-curve story, not a balance story. The Class I versus Class II distinction matters throughout: LME-deliverable Class I (high-purity) availability can tighten while all-form balances show surplus, and the paper's balances do not resolve that split. Battery demand underperformance is the demand-side risk; stainless remains the volume base.

Scenarios give a 2030 base of -120 kt (-2.7 percent): modest deficit, with a range of +294 to -468 kt (Modelled, C). The verdict: recent surplus, comfortable stocks, forecast too unstable to trade on, and a two-tier cost structure that segments the market by origin. What would overturn the loose reading: Indonesian quotas tightening further rather than normalizing, combined exchange stocks falling below about 300 kt (roughly 30 days of cover, against 478 kt and 46.6 days now), or a Class I-specific squeeze visible in LME spreads (the companion paper's evidence (Kilambi Research, 2026b)). What would confirm the tight reading for only part of the market: sustained LME tightness alongside all-form surplus, the Class I/II split made visible.

6.6 Tin: small deficits on the weakest data of the six

Tin's balance was +3.5 kt in 2022, +14.3 kt in 2023, -5.3 kt in 2024, and -5.6 kt forecast for 2025 (Observed, C) (International Tin Association, 2025), but a later ITA article disputes the 2024-2025 deficit characterization and the conflict is unresolved. Balances for 2020-2021 and 2026 were not located and are shown as gaps (Figure 8). Visible stocks were 9.6 kt, 9.2 days of cover (Reconstructed, B) (Kilambi Research, 2026a) (Figure 4): thin in absolute days, on a small market where a single mine restart moves the balance.

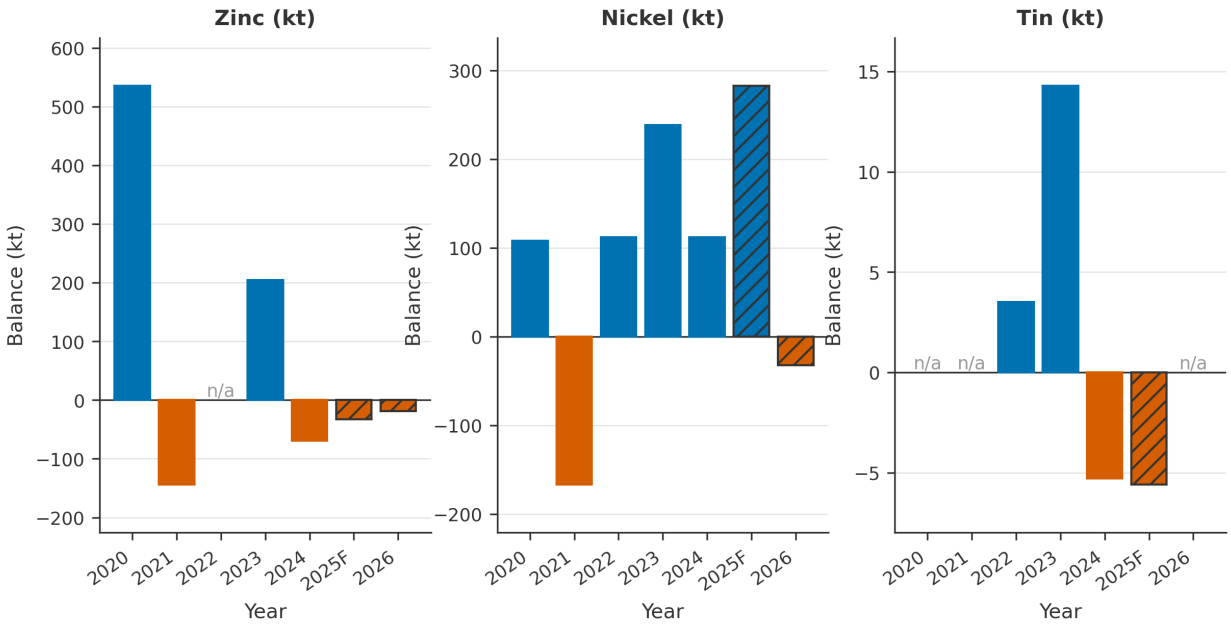


Figure 8. Refined balances 2020-2026 for zinc, nickel, and tin. Hatched bars are forecasts or preliminary values; "n/a" marks years not located (shown as gaps, never interpolated). Sources: ILZSG, INSG, ITA.

Myanmar is the swing supplier through China: Wa State's mine ban removed an estimated 30,000-32,000 t of contained tin (Observed, C), and Myanmar's share of China's tin ore imports has run from 33 percent to over 90 percent across vintages (Observed, B). Yunnan power cuts cost a further 1,000-2,000 t (Observed, B). China refines about 52 percent of world tin against 25 percent of mine output (Reconstructed, A), so the Myanmar-China ore corridor is the physical chokepoint.

The incentive range is the high $20,000s to mid $30,000s per tonne (Reconstructed, C), against spot of $55,385/t in August 2026 (Observed, A) (Figure 9). Spot far above incentive with small deficits and thin stocks would normally be read as a market calling forth supply; whether it is depends on Myanmar policy and permitting timelines the paper cannot verify. Scenarios give a 2030 base of -12 kt (-2.9 percent): modest deficit, range +24 to -44 kt (Modelled, C).

The verdict is hedged harder than for any other metal: small deficits and thin stocks, on data too weak to support a strong conclusion. What would overturn even the mild tight reading: an actual reported surplus, visible stocks rebuilding above 15 kt, a Myanmar restart, or flat consumption with spot below $30,000/t.

6.7 Lead and the platinum-group metals: brief checks

Lead is adequately supplied. The 2025 forecast surplus is +82 kt (Observed, C) (International Lead and Zinc Study Group, 2026), LME stocks sit at 10.1 days (Observed, B), and recycling supplies roughly 60 to 70 percent of refined metal (Reconstructed, B), the highest secondary share in the group. Lead's demand is concentrated in batteries with limited substitution risk in the paper's window. What would change the verdict: sustained ILZSG deficits or evidence that scrap collection, not smelting, has become the binding constraint. Neither is in the data.

Platinum ran a 2025 deficit of 1,082 koz on WPIC numbers and 951 koz on Johnson Matthey numbers (OBSERVED, B) (World Platinum Investment Council, 2026; Johnson Matthey, 2026); the 2026 deficit narrows sharply to 240 koz (WPIC) and 371 koz (JM) (OBSERVED, C). Palladium and rhodium are forecast to move into surplus in 2026 (OBSERVED, C). PGM above-ground stocks are modeled estimates, not measurements (MODELLED, C), and investment demand is the swing category that makes PGM balances the most revision-prone in the set. What would overturn the narrowing-deficit reading: an investment-demand reversal, a recycling recovery, or the forecast 2026 surpluses failing to materialize. The PGMs stay brief here by design; their investment-driven demand dynamics belong with the companion paper's positioning analysis (Kilambi Research, 2026b).

6.8 Across metals: China, incentives, energy, and secondary supply

China's refining dominance is the cross-cutting structural fact (Figure 10). Refined shares run about 49 percent for copper, 47 percent for zinc, 52 percent for tin, 38 percent for lead, and 29 percent for nickel, against mine shares of 8, 34, 25, 44 percent, and a small nickel share respectively (RECONSTRUCTED, A/B). The mine-refined wedge is the measure of the rest of the world's dependence on Chinese processing. It also concentrates policy risk: export licensing, VAT treatment, and SRB stockpiling decisions move physical availability without moving mine output.

The incentive-versus-spot check (Figure 9) sorts the metals into three groups. Copper and zinc: spot above incentive, consistent with adequate-to-loose annual markets. Nickel: spot between the Indonesian cash cost and the Western incentive, a two-tier market where origin determines profitability. Silver and tin: spot well above incentive ranges, which in a frictionless market would call forth supply; silver's by-product structure and tin's policy chokepoint are the named frictions. Gold has no quoted incentive price, a disclosed gap.

The 2030 scenario mapping under Section 5 thresholds is summarized in Table 2. Every balance-testable metal flips sign across the four paths. The base path reads "adequately supplied" for copper, silver, and zinc, "modest deficit" for nickel and tin; no metal reaches "severe structural deficit" or "acute squeeze" on the base path. The labels describe mechanical paths, not forecasts, and the table's message is the width of the ranges, not the center of any one of them.

Table 2. 2030 scenario balances mapped to plain-language labels

Metal	Conservative	Base	High demand	Low supply
Copper	+1,044 kt: supplied	-284 kt: supplied	-1,772 kt: severe	-1,357 kt: modest
Zinc	+743 kt: supplied	-10 kt: supplied	-599 kt: modest	-582 kt: modest
Nickel	+294 kt: supplied	-120 kt: modest	-468 kt: severe	-440 kt: severe
Tin	+24 kt: supplied	-12 kt: modest	-39 kt: severe	-44 kt: acute squeeze
Silver	+86.6 Moz: supplied	-12.5 Moz: supplied	-106.9 Moz: severe	-69.9 Moz: modest

Notes: labels abbreviate adequately supplied / modest deficit / severe structural deficit / acute squeeze; only tin's low-supply path (-10.8 percent) reaches the acute-squeeze band. All paths (MODELLED, C). Gold excluded: balance identically zero under WGC accounting.

Energy sensitivity is a zinc-led risk with copper-smelting spillovers. The 2022 episode showed European power prices converting into physical supply cuts within weeks; the observed EUR 340-527/MWh range is not a threshold but a precedent (OBSERVED, B) (Nyrstar et al., 2022). With Chinese smelting overcapacity now the larger structural feature, the next energy shock would more likely bind through concentrate competition (lower TCs) than through European curtailments.

Secondary supply responded where it could. Silver recycling rose to 197.6 Moz in 2025 on the price (Observed, B); gold recycling reached 1,404.3 t in 2025, up 3 percent (Observed, B); lead's 60-70 percent secondary share is structural (Reconstructed, B). For copper, zinc, nickel, and tin the price-responsiveness of scrap is not well measured, and the paper does not assume a response it cannot see. Nickel's reported recycling rate of about 3 percent (Observed, C) is the clearest statement that secondary supply is not a near-term buffer there.

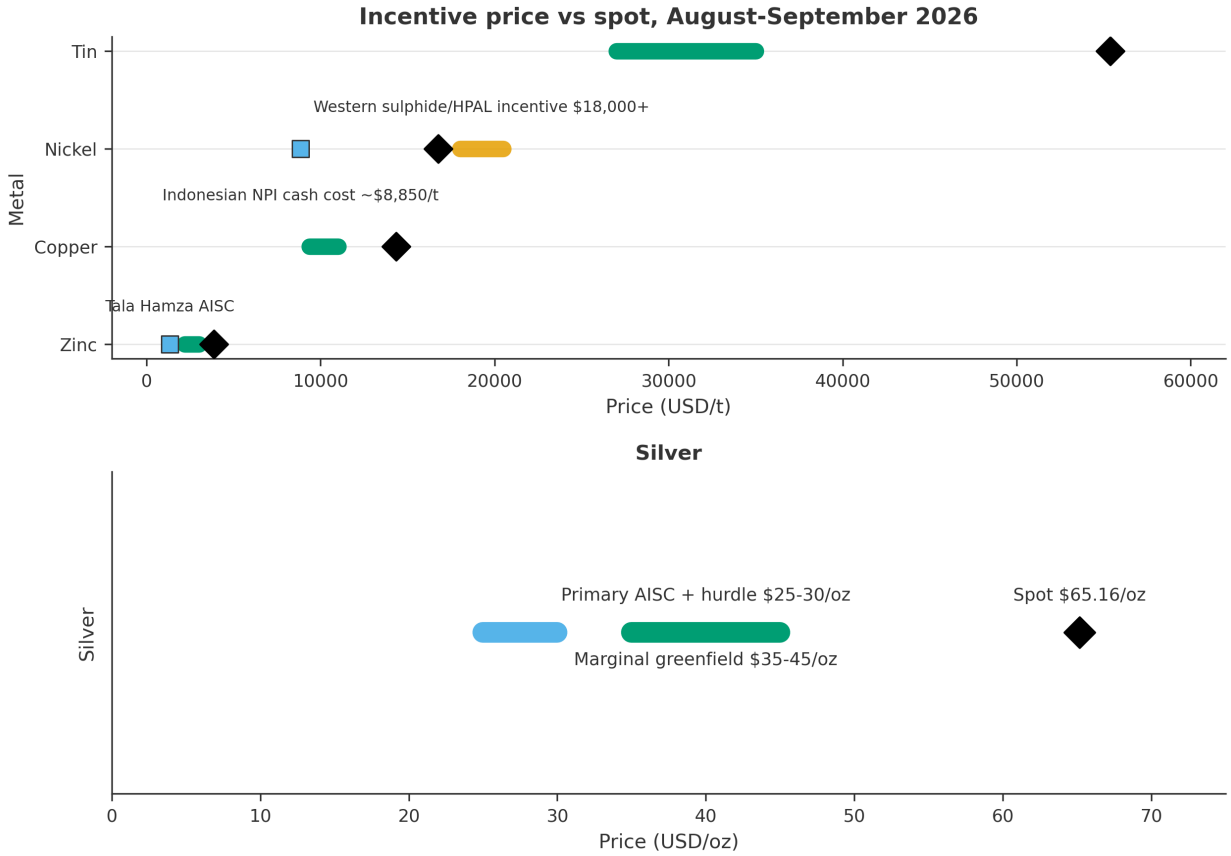


Figure 9. Incentive price versus spot, August-September 2026. Diamond: IMF August 2026 spot. Green bar: incentive band (Reconstructed, C); square: reference cost point. Gold has no defensible incentive price and is excluded, a disclosed gap.

7 Limitations and robustness

7.1 What would overturn each conclusion

Table 3 collects the falsifiers: observable facts that would change the verdicts in Section 6. They are stated with approximate thresholds where the data allow and as directional conditions where they do not. The paper's conclusions are conditional on these facts not obtaining; several are close enough to monitor.

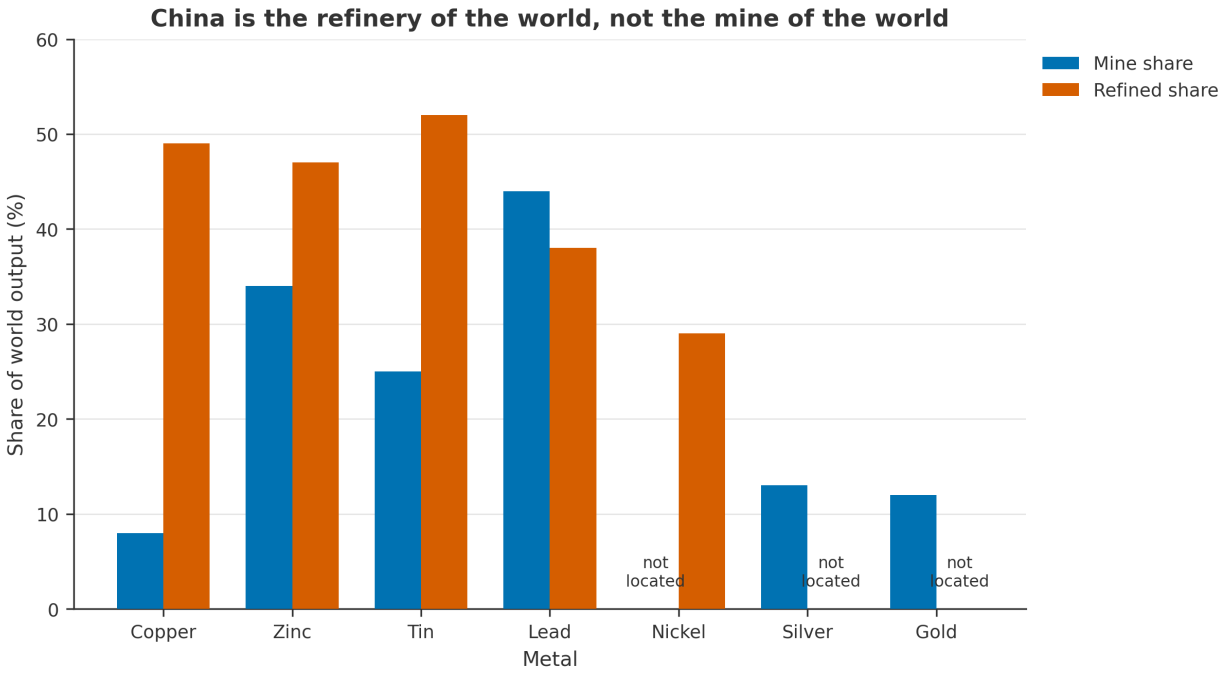


Figure 10. China is the refinery of the world, not the mine of the world. Shares of world mine and refined output by metal; "not located" marks cells with no verified figure. Sources: study groups, USGS, China customs reporting.

7.2 Robustness in both directions

The paper checks its conclusions against adverse evidence as well as supportive evidence. For copper, the adverse facts are on the table: reserves growing, mine utilization falling, spot above incentive, and the 2025 surplus itself, all of which argue against scarcity and are reported in Section 6.1 rather than buried. For silver, the adverse fact is the 2025 forecast revision (-117.6 to -40.3 Moz), which shrinks the deficit narrative the paper's own history would otherwise support. For nickel, the adverse fact is the 293 kt six-month revision of the 2026 forecast, which is why the paper refuses to trade on it.

Forecast revision magnitudes are themselves a finding about stability. Copper's 2024 balance was revised from 301 kt to 69 kt; silver's 2025 deficit from -117.6 to -40.3 Moz; nickel's 2026 balance from +261 to -32 kt. Revisions of this size relative to the balances mean that any conclusion resting on a single vintage is fragile. The paper's verdicts rest on multi-year patterns and stock levels, which survive revision better than any single annual figure, but the fragility is disclosed, not assumed away.

The scenarios hold up in the sense that matters: their signs flip for every metal. A conclusion that required selecting the high-demand path for copper while selecting the conservative path for zinc would be cherry-picking; the paper selects no path. The both-directions check on scenarios is that the conservative path shows surpluses for all five balance-testable metals, which falsifies any claim that the model structure builds in scarcity.

Table 3. Falsifiers: the facts that would overturn each metal's verdict

Metal	Falsifying facts
Copper	Third consecutive refined surplus would confirm looseness; overturning facts: LME on-warrant above 200 kt with contango, term TC/RC above about $80/dmt, or project supply replacing depletion two years running.
Silver	Two consecutive surpluses with spot below $35/oz; COMEX registered above 150 Moz; vault free float above 40%; PV demand below about 130 Moz.
Gold	Central-bank buying below 500 t for two years; mine growth above 2% for three years; recycling above 1,600 t. (No scarcity claim is made, so these falsify flow-tightness, not the paper.)
Zinc	ILZSG reported stocks above 850 kt; LME above 200 kt; European smelter restarts completing; concentrate TCs recovering materially.
Nickel	Indonesian quotas tightening further; combined exchange stocks below about 300 kt (roughly 30 days of cover); battery-demand outperformance. Class I versus Class II must be distinguished throughout.
Tin	Actual reported surplus; visible stocks above 15 kt; Myanmar restart; flat consumption with spot below $30,000/t.
Lead	Sustained ILZSG deficits; scrap collection becoming the binding constraint.
PGMs	Investment-demand reversal; recycling recovery; forecast 2026 surpluses failing to materialize.

7.3 Known gaps

No defensible gold incentive price was found; none is quoted. The full World Silver Survey 2026 table and methodology were not retrieved, so silver demand components rest on summary reporting. Tin balances for 2020-2021 and 2026 were not located. Current Chinese bonded copper stocks are unknown. Copper, zinc, nickel, and tin end-use splits are weak reconstructions. Copper's 2026 forecast is stale, predating the 2025 actual. The copper depletion estimate is (Modelled, C) on unretrieved underlying data. Several project capacities and probability weights are secondary or assumed. The tin 2024-2025 balance conflict is unresolved. The zinc 2024 relay conflict (-69 versus -62 kt) is disclosed. Lead's USGS mine figures came from a search snippet because the direct PDF was not retrieved. PGM above-ground stocks are modeled estimates. Zinc's EUR 340-527/MWh curtailment range is observed, not a break-even threshold. Global recycling shares and response thresholds are unknown for several base metals. Inventory snapshots mix dates and some pools overlap. Gold's component accounting has a 2024 reconciliation gap. Silver's multi-year reported deficits without an earlier price break remain economically unresolved. Each gap is marked in the text where it binds; none is filled by assumption.

8 Conclusion

Physical metals markets in 2025-2026 do not present a uniform scarcity picture. Copper's annual refined market is loose while its concentrate market is historically tight; the two facts coexist because smelting overcapacity sits between them. Silver's deficits are persistent but revised smaller and

absorbed by stocks, with a supply structure that mutes the price signal. Gold's balance is zero by construction and its price clears through a 60-year stock. Zinc is close to balanced with upstream fragility. Nickel is coming off surpluses into an unstable forecast. Tin is thin on thin data. Lead is adequately supplied; the PGMs are narrowing.

The common threads are structural rather than cyclical: China's refining share, the by-product trap in silver, the two-tier cost curve in nickel, the policy chokepoint in tin ore, and the energy sensitivity of smelting. The 2030 scenarios show that small changes in growth assumptions flip the sign of every metal's balance, which is an argument for humility about long-horizon tightness claims and for the falsifier discipline this paper adopts: each verdict comes with the facts that would overturn it, stated in advance.

The companion paper, Kilambi Research (2026), "Do Paper Markets Price Physical Scarcity?", KR-WP-2026-14, takes up the other half of the question: whether futures curves, positioning, and delivery mechanics price the physical conditions documented here.

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10 Data appendix

The machine-readable dataset behind every figure and table lives in `metals-research/data/` and is inventoried here with row counts as of September 2026. Every row carries source, as-of date, access date, and evidence grade.

File	Rows	Contents
`metal_balances.csv`	168	Annual balances 2020-2026 plus 2030/2035 scenarios, all metals
`metal_inventories.csv`	30	Exchange and reported stocks with days of cover
`china_metals.csv`	50	China mine, refining, consumption shares; trade; SRB
`china_policy_timeline.csv`	30	China metals policy timeline
`copper_mine_supply.csv`	29	Copper mine supply by country
`silver_mine_supply.csv`	20	Silver mine supply by country
`gold_mine_supply.csv`	20	Gold mine supply by country
`nickel_mine_supply.csv`	15	Nickel mine supply by country
`tin_mine_supply.csv`	15	Tin mine supply by country
`zinc_mine_supply.csv`	15	Zinc mine supply by country
`gold_demand_components.csv`	45	Gold demand by component
`metals_demand_by_endmarket.csv`	89	Base-metal demand by end market
`demand_growth_drivers.csv`	12	Demand growth driver assumptions
`silver_recycling_stocks.csv`	20	Silver recycling and stocks
`copper_projects.csv`	29	Named copper projects with probability weights
`nickel_projects.csv`	8	Named nickel projects with probability weights
`tin_projects.csv`	8	Named tin projects with probability weights
`zinc_projects.csv`	8	Named zinc projects with probability weights
`copper_concentrate_tc_rc.csv`	14	Term and spot TC/RC series
`raw_M.W00.*.csv`	440 ea	IMF monthly spot price series, six metals
`KEY_NUMBERS.md`	–	Canonical headline values with provenance
`SHARED_FALSIFIERS.md`	–	Per-metal falsifiers used in Table 3

Vintages differ across files by construction: balances run on study-group publication calendars (February to April for prior-year actuals), exchange stocks are September 2026 snapshots, and prices are August 2026 monthly averages. Preliminary values are labeled preliminary in the CSVs and in the text. Where sources conflict (zinc 2024, tin 2024-2025), both values are kept with provenance and the conflict is disclosed in the text rather than resolved by selection.